Milestone 1: Core Improvements & Bug Fixes - Delivery Report

Date: 9 February 2026

Milestone: Milestone 1 - Core Improvements & Bug Fixes

Status: ✅ Complete

Preview URL: science-canvas-creator.vercel.app

Executive Summary

All seven deliverables from Milestone 1 have been completed and verified:

- Newly uploaded icons and existing icons render correctly on canvas
- Canvas size templates now apply dimensions accurately
- Paginated icon display shows 20 icons per page with navigation controls
- Enhanced search finds icons across both names and tags with partial word matching
- Admin tag management system enables bulk tagging via CSV import/export and inline editing
- Image crop tool provides five crop modes with non-destructive editing
- Canvas performance optimization enables smooth handling of 100+ icons with projects saving successfully without size limit errors.

All deliverables include comprehensive test coverage as specified in the milestone requirements. Additional automated tests were included beyond the initial milestone scope in order to ensure features were completed successfully All tests are 100% passing.

Deliverables Completed

Canvas Size Template Fix

What Was Delivered:

Canvas size templates now function correctly. When a preset paper size is selected (A4, US Letter, Nature journal formats) or custom dimensions are entered, the canvas resizes to the

exact specified dimensions. The rulers and grid update automatically to match the new canvas size, and dimensions persist correctly through save/reload cycles.

How It Was Verified:

Three automated end-to-end tests verify template functionality:

- A4 Portrait template sets correct dimensions (2480 × 3508 px) with proper portrait aspect ratio
- Letter Landscape template sets correct dimensions (3300 × 2550 px) with proper landscape aspect ratio
- Template selection persists correctly after project save and reload (verified with Nature Single Column template)

Manual testing verified additional scenarios:

- Various preset paper sizes (A4, Letter, Nature journal formats) resize canvas to exact dimensions
- Custom dimensions (1200×800px, 1024×1024px) are applied exactly as specified without auto-scaling
- Rulers display correct canvas width and height after dimension changes
- All template sizes persist correctly through save/reload cycles

SVG Upload Fix

What Was Delivered:

Newly uploaded SVG icons now render correctly on the canvas. The system properly processes both single SVG uploads and bulk ZIP uploads containing multiple SVG files. Icons appear in the icon library after upload and render correctly when clicked or dragged onto the canvas, displaying with proper colors and sizing.

A migration script was created to retroactively fix existing icons that were uploaded before this fix. The script applies the same `minifySVG()` function to remove XML declarations, metadata, and editor-specific attributes that can cause rendering issues. Before running the migration on production, all 6,341 icons (725 MB) were backed up to ensure data safety. The migration successfully cleaned 1,331 icons in production and 1,148 icons in development, saving approximately 55 MB per database.

How It Was Verified:

Two automated end-to-end tests validate the complete upload-to-render pipeline:

- Single SVG upload: Admin uploads SVG file → User searches and finds icon → User clicks to add to canvas → Icon renders correctly at cursor position
- Bulk ZIP upload: Admin uploads ZIP with multiple SVGs → User searches and finds icon → User drags to canvas → Icon renders correctly at drop position

Paginated Icon Display

What Was Delivered:

Icon library now displays 20 icons per page with Previous/Next navigation buttons and a page indicator showing "X / Y" format. Pagination state is preserved when switching between categories, so returning to a previously viewed category shows the same page you were on. Categories with more than 20 icons display only 20 initially, requiring navigation to see additional pages.

How It Was Verified:

Six automated end-to-end tests verify complete pagination functionality:

- Icon categories with >20 icons display only first 20 icons initially
- Next button loads page 2 with next 20 icons and updates page indicator
- Previous button returns to page 1 correctly
- Page indicator displays current page and total pages accurately (e.g., "2 / 5")
- Switching to a different category resets pagination to page 1
- Returning to a previously viewed category preserves the page you were on

Enhanced Search with Tagging

What Was Delivered:

Icon search now works across both icon names and tags with partial word matching. Search results are sorted by relevance (exact name matches rank higher than tag matches, which rank higher than partial matches). Results appear in a unified view showing icons from all categories. Clearing the search returns to category browsing mode, and searches with no results display a "No icons found" message.

How It Was Verified:

Six automated end-to-end tests plus five unit tests verify complete search functionality:

End-to-End Tests:

- Searching for exact icon name "mitochondria" returns matching results

- Searching for partial word "mito" successfully finds "mitochondria"
- Searching for tag "cellular-organelle" returns all icons with that tag
- Search results display icons from multiple categories in unified view
- Clearing search successfully returns to category browsing mode
- Search with no results displays "No icons found" message

Unit Tests:

- Exact name matches rank higher than tag matches
- Partial word matching works correctly (substring search)
- Search is case-insensitive
- Search handles special characters correctly
- Empty search returns no results

Admin Tag Management System

What Was Delivered:

Administrators can now manage icon tags through three methods: CSV export downloads all icons with their current tags, CSV import enables bulk tag updates with validation and preview before applying changes, and inline editing allows quick tag updates directly in the admin interface. Tag auto-suggestions help maintain consistency by showing existing tags. Batch progress notifications keep administrators informed during bulk updates.

How It Was Verified:

Six automated end-to-end tests verify complete CSV tagging workflow:

- CSV export downloads file with correct format (headers include id, name, category, tag columns)
- CSV import successfully updates database with modified tags
- Invalid CSV format shows appropriate error messages and disables import button
- Preview dialog displays accurate update counts before import is executed
- Success message displays correct count of updated icons after import
- Tags appear in icon search results immediately after CSV import completes

Manual testing verified additional functionality:

- CSV export successfully downloads complete icon database with current tags (6,431 icons)
- CSV import with 6,355 tag updates processed successfully with 98.3% match rate
- Inline tag editing updates are immediately visible in both admin interface and search results

- Tag auto-suggestions display existing tags correctly
- Progress notifications show current batch status during bulk updates

Image Crop Tool

What Was Delivered:

An image cropping UX with five modes: Rectangle, Square (1:1 aspect ratio locked), Ellipse, Circle (1:1 aspect ratio locked), and Free-form polygon. The crop tool opens in a full-screen lightbox overlay with translucent background showing the area to be removed. Cropping is non-destructive—the original image is preserved, and cropped images can be edited or reset to original at any time.

When an uncropped image is selected, five mode buttons appear in the Properties Panel. After cropping, Edit Crop and Reset to Original buttons appear instead. Edit Crop restores the original uncropped image with the crop overlay at its exact previous position for adjustments. Free-form mode provides draggable polygon anchor points with the ability to add or remove points (minimum 3 points). Crop state persists through save/load cycles and works correctly with undo/redo.

How It Was Verified:

Three automated unit tests verify crop calculations:

- Polygon crop path calculation produces valid clip paths with correct positioning
- Circle crop maintains perfect 1:1 aspect ratio during resize operations
- Square crop maintains perfect 1:1 aspect ratio during resize from any corner

Seven automated end-to-end tests verify complete crop workflow:

- Free-form crop opens lightbox with draggable anchor points
- Dragging anchor points updates crop preview in real-time
- Apply crop creates cropped image with custom shape
- Edit Crop restores original image with previous crop overlay position
- Reset to Original removes crop and restores full image
- Circle crop displays resize handles and maintains aspect ratio
- Square crop maintains 1:1 aspect lock during resize

Manual testing verified:

- All five crop modes accessible from Properties Panel when image selected
- Crop state persists correctly through save/load cycles
- Undo/redo operations work correctly for all crop operations

Canvas Performance Optimization

What Was Delivered:

Canvas performance has been significantly improved to handle large projects with many icons. The application now handles 100+ icons on canvas with minimal lag, and projects save successfully without hitting the previous 4MB size limit error.

Icon storage efficiency:

Icons are now saved using reference values (icon ID, position, dimensions, styles), drastically reducing project size. These values are used to load the icons on project load - this results in a longer loading time (a progress bar was added). The longer load time is a trade off that allows large projects to be saved and loaded efficiently.

Icon render efficiency:

- Alignment and spacing guide lines were optimized to reduce the maximum number of guides rendering at once. Guide lines are also disabled when over 250 icons are on the canvas.
- Membranes are stored with all icon details, but are displayed on the canvas as images to prevent rendering many icons for each membrane
- History management was refactored to store transformations rather than serialized object data.
- Autosave was optimized to only save following a delay after canvas edits are completed.

Note: Some very comlex icons still may result in some lag because these can be very data-intensive to render. Performance can be further improved by rendering the icons as images, but this will reduce the quality.

How It Was Verified:

Manual testing verified all performance requirements from the milestone specification:

- Canvas with 100 icons maintains smooth object manipulation without noticeable lag
- Projects with 100+ icons save successfully without "PROJECT TOO LARGE TO SAVE" error
- History save operations complete in under 100ms (previously 500-5000ms)
- Dragging objects feels responsive with no lag or stuttering
- Project thumbnails generate correctly for projects with many icons
- Undo/redo operations respond immediately to single keypresses
- Performance diagnostics confirm 71 FPS maintained, 33ms load time, 0 idle renders

- Data compression successfully reduces project file size without losing fidelity

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Deliverables Status

Completed:

- ✓ Canvas Size Template Fix (3 E2E tests + manual verification)
- ✓ SVG Upload Fix (2 E2E tests)
- ✓ Paginated Icon Display (6 E2E tests)
- ✓ Enhanced Search with Tagging (5 unit tests + 6 E2E tests)
- ✓ Admin Tag Management System (6 E2E tests + manual verification)
- ✓ Image Crop Tool (3 unit tests + 7 E2E tests)
- ✓ Canvas Performance Optimization (68 unit tests + manual verification)
